

Dynamics test(12 marks)

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1 Blocks(12 marks)

1.1 Question 1(4 marks)

The diagram shows 2 blocks in contact on a smooth surface. A 48N force acts on block 1, and the surface can be considered frictionless. Calculate:

- (a) The acceleration of the system
- (b) The acceleration of each block
- (c) The net force on each block
- (d) The contact forces for each block

2 Section 2(8 marks)

This is another sample set.

Behold, nested questions.

1. For each of the following, solve by completing the square(4 marks)
 - (a) pretend i put a quadratic here
 - (b) pretend i put a quadratic here
 - (c) pretend i put a quadratic here
 - (d) pretend i put a quadratic here
2. For each of the following, solve using the quadratic formula(4 marks)
 - (a) pretend i put a quadratic here
 - (b) pretend i put a quadratic here
 - (c) pretend i put a quadratic here
 - (d) pretend i put a quadratic here